

M.Tech Project Work Distribution

Programme: M.Tech (Artificial Intelligence & Machine Learning) **Institute:** Indian Institute of Technology Jodhpur **Project Duration:** Two Semesters (Project-I + Project-II)

Credits Distribution:

Semester	Credits
Semester I (Project-I)	5 Credits
Semester II (Project-II)	11 Credits

Project Title: *Ontology-Guided Multilingual Semantic Retrieval and Explainable AI Framework for Classical Ayurvedic Knowledge Systems*

Student: Manvendra

Semester I – Project-I (5 Credits)

Task Title

Design and Evaluation of a Multilingual OCR Pipeline for Classical Ayurvedic Texts

1. Objective

To design, implement, and rigorously evaluate an OCR-based digitization pipeline capable of converting scanned Sanskrit Ayurvedic texts into clean, structured, machine-readable UTF-8 encoded data suitable for downstream natural language processing tasks.

2. Scope of Work

- Collection and preparation of scanned Sanskrit Ayurvedic texts.
- Implementation of OCR pipeline for text extraction.
- Encoding normalization and text cleaning (UTF-8 standardization).
- Removal of OCR artifacts and structural noise.
- Preservation of textual hierarchy (grantha, chapter, verse).

- Development of automated evaluation scripts for:
 - Character Error Rate (CER)
 - Word Error Rate (WER)
 - Comparative performance analysis of OCR configurations.
 - Error categorization and qualitative analysis (diacritics, sandhi, compound words).
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3. Deliverables

- Clean and structured Sanskrit digital corpus.
 - OCR evaluation report with quantitative metrics.
 - Error analysis documentation.
 - Reproducible preprocessing pipeline.
 - Contribution to Semester I project report.
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4. Academic Contribution

This task establishes a reliable digital foundation for classical Ayurvedic knowledge representation, ensuring high-quality textual data for semantic modeling and ontology construction in subsequent stages.

Semester II – Project-II (11 Credits)

Task Title

Development of an Ontology-Guided Retrieval-Augmented Generation System for Explainable Disease-to-Treatment Mapping

1. Objective

To design and implement a hybrid neuro-symbolic AI framework integrating vector-based semantic retrieval, ontology-guided filtering, and large language models (LLMs) to generate grounded, citation-backed, and explainable disease-to-treatment recommendations from classical Ayurvedic texts.

2. Scope of Work

- Generation of multilingual sentence embeddings.
 - Implementation of vector-based semantic search system.
 - Integration of ontology-guided filtering using the constructed knowledge graph.
 - Design of hybrid retrieval strategy (vector + graph-based constraints).
 - Development of Retrieval-Augmented Generation (RAG) pipeline.
 - Integration with a Large Language Model for:
 - Disease-to-treatment inference
 - Structured explanation generation
 - Source citation extraction (grantha, chapter, verse)
 - Implementation of faithfulness control mechanisms to reduce hallucination.
 - Comparative experimental evaluation of:
 - Vector-only retrieval
 - Graph-only retrieval
 - Hybrid ontology-guided retrieval
 - Quantitative evaluation using Precision@K, Recall@K, MAP, and NDCG.
 - Qualitative evaluation of explainability and citation faithfulness.
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3. Deliverables

- Hybrid AI system prototype.
 - Experimental evaluation results and comparative analysis.
 - Citation-backed explainable output framework.
 - Contribution to final thesis and research publication draft.
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4. Academic Contribution

This work contributes a novel ontology-guided neuro-symbolic retrieval framework for classical Ayurvedic knowledge systems, advancing research in explainable medical AI and multilingual semantic reasoning.

Indian Institute of Technology Jodhpur M.Tech – Artificial Intelligence & Machine Learning